

BlackBerry 10

Android APK Compatibility Repository

Structured guide to sideloading Android APKs on BB OS 10.3.x · API 18 runtime

Categories: Web Browser · Offline Media · Utility · Social

Includes version numbers, archive sources & troubleshooting checklist

Runtime Constraints

Constraint	Detail
Max Android API	18 (Android 4.3 Jelly Bean MR2)
Runtime Engine	Dalvik (not ART) — pre-compiled APK bytecode
Multi-core Limit	Apps capped at 2 CPU cores (even on Passport quad-core)
Google Play Services	Not supported — GMS-dependent apps fail at login or runtime
Adobe AIR Apps	Supported through BB OS 10.3.1 only
BAR Signing	Signing servers offline since Jan 2022 — Developer Mode required for unsigned BAR
Preferred Arch	armeabi-v7a (ARM 32-bit) — always prefer over arm64 or x86_64
Direct APK Install	Requires BB OS 10.2.1+; BB OS 10.3.x recommended for API 18 cap

Category 1 — Web Browsers

App	Version	Min API	Source	Notes
Opera Mini	7.6.4	API 3	APKMirror	Best-tested browser; confirmed on Z10, Q10, Classic. Use armeabi-v7a build.
Fennec F-Droid (Firefox)	68.11.0 / 68.12.0	API 16	APKMirror Mozilla Archive	Last version supporting Android 4.1–4.4. 68.x is the community-confirmed BB10 ceiling.
BEOL Browser	1.8.1	API 4.x	GitHub Direct APK	Built specifically for BB10 Android runtime. Minimal footprint, ARM-optimized.
Kiwi Browser	Legacy builds	API 16	APKMirror	Chromium-based; community-reported working. Verify minAPI before download.
UC Browser	Pre-2018 builds	API 14	APKPure	Earlier builds confirmed via Amazon Appstore on BB10. Avoid 2019+ versions.

NOTE: Firefox 79+ (Fenix/GeckoView) requires API 21 — will fail with error -12. Chrome requires API 21+. For Discord: no APK works; use mobile web via Fennec 68.

Category 2 — Offline Media

App	Version	Min API	Source	Notes
VLC for Android	Any 3.x	API 17	APKMirror Uptodown	Universally confirmed. Handles MKV, MP4, AVI, FLAC, MP3. Use armeabi-v7a.
Moon+ Reader	7.3 (API 16 build)	API 16	APKMirror (API 16)	Best e-book reader for BB10. Supports ePub, MOBI, PDF. v7.3 = last API 16 build.
NewPipe Legacy Revo	Latest available	API 16	GitHub – ShareASmile	YouTube without GMS. Fork of NewPipe Legacy for API 16+ devices. Confirmed on BB10.
SkyTube Legacy	Latest available	API 16	GitHub – SkyTube	YouTube client; flagged working by Project Berry 2.0 community.
MX Player Classic	Pre-1.10	API 14	APKMirror	HW decode mode recommended. Avoid 2019+ builds (require API 21+).
Spotify Lite	1.5.46.7	API 18	APKMirror	Confirmed on Q10. Full Spotify 8.5.85.894 also reported. No GMS for core playback.
ZArchiver	1.0.9 / 1.0.10 armeabi-v7a	API 14	APKMirror ZDevs Official	ZIP, 7z, RAR, TAR support. Get armeabi-v7a build, not arm64.

TIP: For IPTV apps: set decoder to HW (Hardware) mode to reduce crashes. VLC SW (Software) mode is fallback for codec issues.

Category 3 — Utility

App	Version	Min API	Source	Notes
F-Droid	1.12.1	API 14	F-Droid Archive	Last version supporting Android 4.2+. Acts as FOSS app store on BB10.
Magic Earth Navigation	7.x	API 16	APKMirror	Offline GPS maps, no GMS required. Confirmed on Passport and Classic.
MicroG / UnifiedNlp	Latest stable	API 4.x	MicroG Project	GMS stub. Enables network location + some account auth without Google.
Helium Backup	Legacy	API 14	APKMirror	App backup without root. Reported working on BB10 Q10.
ES File Explorer	3.x legacy	API 8	APKMirror	Pre-4.x builds avoid adware. Classic file manager, confirmed on BB10.
Term48 (Native)	Latest BAR	BB10 Native	CrackBerry Forums	Native BB10 terminal emulator. Pairs with SSH/networking tools.
Speedtest by Ookla	Legacy	API 14	APKMirror	Network diagnostics. Older builds confirmed working.

App	Version	Min API	Source	Notes
RAR	Legacy	API 14	APKMirror	Official RARLAB tool. Older builds work on API 18.

Category 4 — Social / Communication

App	Version	Min API	Source	Notes
Telegram X	0.25.6.1615 (universal / armeabi-v7a)	API 16	APKMirror	Use armeabi-v7a or universal API 16+ build. Newer 6.0+ builds will fail.
RedReader	Latest (verify minAPI)	API 16	F-Droid APKMirror	Open-source Reddit client, no GMS needed. Recent builds may need API 21 — verify.
Xtra (Twitch)	Legacy builds	API 16	GitHub – Xtra	Unofficial Twitch client. Initial login requires Android browser on BB10.
SoundHound	Legacy	API 14	APKMirror	Music identification. Works without GMS. Shazam requires newer API.
Facebook Lite	v1.x – v50.x range	API 9	APKMirror	Pre-2018 builds work. Newer versions require API 21+.
Spotify Lite	1.5.46.7	API 18	APKMirror	Confirmed on Q10. GMS not required for core streaming.

NOT COMPATIBLE: WhatsApp (all current versions), LinkedIn, Facebook Messenger: dropped API 18 support. No workaround available.

Sideloading Methods

Method 1 — Direct APK (Recommended, BB OS 10.2.1+)

1. Enable: Settings → App Manager → Installing Apps → Allow apps from other sources
2. Copy APK to SD card or internal storage
3. Open File Manager → navigate to APK → tap to install
4. Accept permissions prompt (all permissions are granted at install-time; cannot be modified post-install)

Method 2 — Darcy's BlackBerry Tools (DBBT) via PC

1. Download DBBT (Windows, requires .NET Runtime) — search CrackBerry forums for current link
2. Enable Developer Mode: Settings → Security and Privacy → Developer Mode → set password
3. Connect BB10 via USB; note the Development IP Address displayed on device
4. Launch DBBT; drag and drop APK into the install window

Method 3 — Chrome Extension (PlayBook App Manager, legacy)

1. Install the PlayBook App Manager Chrome extension (legacy Chrome versions only)
2. Enable Developer Mode on device and note the IP address
3. Enter IP in extension; drag BAR or APK files to the extension window to install

Method 4 — BAR File Conversion

1. BlackBerry signing servers shut down January 2022 — BAR files are now unsigned
2. Use [APK-to-BAR Converter](#) to generate unsigned BAR
3. Deploy via DBBT or Chrome Extension with Developer Mode active on device

Troubleshooting Checklist

Pre-Install Verification

Check	Action / Detail
[] Check APK API level	Must be ≤ 18 . On PC: <code>aapt dump badging file.apk grep sdkVersion</code>
[] Verify architecture	Download armeabi-v7a builds. arm64-v8a and x86_64 will not install correctly on BB10.
[] Confirm BB OS version	Must be 10.2.1+ for direct APK. Must be 10.3.x for API 18 apps.
[] Enable third-party installs	Settings → App Manager → Installing Apps → ON (off by default).
[] Check free storage	Large apps (>50 MB) may fail with -4 if internal storage is under ~100 MB free.

Error Code Reference

Code	Name	Cause	Fix
-4	INSTALL_FAILED_INSUFFICIENT_STORAGE	Not enough free storage space	Clear app cache; free internal storage; use SD card
-12	INSTALL_FAILED_OLDER_SDK	APK requires API > 18	Download an older APK version targeting API ≤ 18
-25	INSTALL_FAILED_VERSION_DOWNGRADE	Trying to install older version over newer	Uninstall current version first, then reinstall
-102	INSTALL_PARSE_FAILED_UNEXPECTED_EXCEPTION	Corrupt APK or parser error	Re-download APK; verify file integrity (MD5/SHA)
-103	INSTALL_PARSE_FAILED_NO_CERTIFICATES	APK has no signing certificate (debug build)	Obtain a properly signed release APK
-104	INSTALL_PARSE_FAILED_INCONSISTENT_CERTIFICATES	Certificate mismatch (different source than installed copy)	Uninstall old version before installing from new source

Source: [BlackBerry Developer Blog — Android Install Error Codes \(2015\)](#)

Runtime / Launch Issues

Symptom	Resolution
[!] App crashes immediately on launch	Likely requires Google Play Services (GMS). Find a GMS-free fork — e.g., Fennec instead of Chrome, NewPipe instead of YouTube.
[!] App installs but shows blank white screen	WebView compatibility issue. Try a different build of the same app targeting API 16–17.

Symptom	Resolution
[!] "Google Play is not supported on your device"	App checks for GMS at runtime. Look for a patched APK or GMS-free fork.
[!] App installs but camera/microphone won't work	Permissions on BB10 are granted at install-time and cannot be changed post-install. Reinstall and accept all permissions.
[!] Location not working in Android app	Install MicroG/UnifiedNlp to provide a GMS-free location backend.
[!] IPTV/video player crashes during playback	Switch decoder to HW (Hardware) in player settings; if that fails, try SW (Software) mode.
[!] App is sluggish / stutters	BB10 limits Android apps to 2 CPU cores. Reduce quality settings in-app. Performance-heavy apps will underperform.
[!] BAR file won't install	Signing servers offline since 2022. Developer Mode must be active on device for unsigned BAR files.

Permission-Specific Behavior

Scenario	BB10 Behavior	Workaround
SMS access	Android apps cannot access BB10 SMS/Hub messages	Use a native BB10 SMS companion app + local socket bridge
Contacts access	Can only read phone-partition contacts (not work partition)	Store contacts in phone partition, not work partition
Background running	"Run When Minimized" is the only toggleable permission	Settings → Security → App Permissions → enable per app
Storage access	Sandboxed; shared SD card media access works	Copy files to SD card and open from there in the app
Post-install permissions	Cannot revoke or grant permissions after install (unlike Android 6+)	Uninstall and reinstall if permission prompt was missed

Quick-Reference Compatibility Matrix

App	Category	Min API	BB10 Status	GMS?
Opera Mini 7.6.4	Browser	API 3	✓ Confirmed	No
Fennec F-Droid 68.12.0	Browser	API 16	✓ Confirmed	No
BEOL Browser 1.8.1	Browser	API 4.x	✓ Confirmed	No
VLC 3.x	Media	API 17	✓ Confirmed	No
Moon+ Reader 7.3	Media	API 16	✓ Confirmed	No
NewPipe Legacy Revo	Media	API 16	✓ Confirmed	No
SkyTube Legacy	Media	API 16	✓ Confirmed	No

App	Category	Min API	BB10 Status	GMS?
ZArchiver 1.0.10 armeabi	Utility	API 14	✓ Confirmed	No
F-Droid 1.12.1	Utility	API 14	✓ Confirmed	No
Magic Earth	Utility	API 16	✓ Confirmed	No
MX Player (pre-1.10)	Media	API 14	✓ Confirmed	No
Telegram X 0.25.6 (armeabi)	Social	API 16	✓ Confirmed	No
Spotify Lite 1.5.46.7	Social/Media	API 18	✓ Confirmed	No
SoundHound (legacy)	Social	API 14	✓ Confirmed	No
Facebook Lite (pre-2018)	Social	API 9	⚠ Older versions only	Partial
RedReader	Social	API 16	⚠ Verify minAPI per build	No
Xtra (Twitch)	Social	API 16	⚠ Login via browser needed	No
WhatsApp (any current)	Social	API 21+	✗ Not compatible	Yes
Firefox 79+ / Fenix	Browser	API 21+	✗ Not compatible	No
Chrome (any)	Browser	API 21+	✗ Not compatible	Yes
LinkedIn (current)	Social	API 21+	✗ Not compatible	Yes
Microsoft Office	Utility	API 23+	✗ Not compatible	Partial

Curated Archive Sources

Source	URL	Best Used For
APKMirror	apkmirror.com	Version-specific APKs; filter by minAPI; verified signed builds
F-Droid Archive	f-droid.org/archive	FOSS apps; 1.12.1 client supports Android 4.2+
IzzyOnDroid F-Droid Repo	apt.izzysoft.de	Extended FOSS archive including Fennec and legacy Privacy Browser builds
Mozilla Archive	archive.mozilla.org	Official Fennec/Firefox release archive back to early versions
ZDevs (ZArchiver Official)	zdevs.ru	All ZArchiver versions by architecture; armeabi-v7a direct download
CrackBerry Forums	forums.crackberry.com	Community-maintained thread of compatible Android APKs for BB10
r/blackberry	reddit.com/r/blackberry	Active BB10 community; 2024 "Ultimate List" thread is a key reference
r/BlackberryPhoenix	reddit.com/r/BlackberryPhoenix	Focused BB10 revival community; browser threads and APK testing logs
Project Berry 2.0 (YouTube)	youtube.com/@ProjectBerry2.0	Video guides for specific APK installs on BB10; active 2024–2025
Uptodown	uptodown.com	Version history with minAPI filters; browse Android 4.1–4.3 builds
APKCombo	apkcombo.com	Filter by Android version directly; find API 16–18 builds

Source	URL	Best Used For
MicroG Project	microg.org	GMS stub for location and account support without full Google services
APK-to-BAR Converter	mobiles24.co/apk2bar	Convert APK to unsigned BAR; requires Developer Mode on device since Jan 2022

Sources: [BlackBerry Developer Blog – Android Install Error Codes](#) · [Amazon Developer Docs – Android on BlackBerry 10](#) · [Wikipedia – BlackBerry 10](#) · [r/blackberry – Ultimate List 2024](#) · [r/BlackberryPhoenix – Working Browsers 2023](#) · [APKMirror](#) · [CrackBerry Forums](#)